

ALFAAH61778

## Tri(1-aziridinyl)phosphine sulfide

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                                                       |                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>产品说明:</b><br><b>Product Description:</b>           | 三(1-吖丙啶基)硫代磷酰<br>Tri(1-aziridinyl)phosphine sulfide                                                                                                                                                                                                                |
| <b>Cat No. :</b>                                      | <b>H61778</b>                                                                                                                                                                                                                                                      |
| <b>Synonyms</b>                                       | Thio-TEPA; Tris(1-aziridinyl)phosphine sulfide                                                                                                                                                                                                                     |
| <b>CAS No</b>                                         | 52-24-4                                                                                                                                                                                                                                                            |
| <b>Molecular Formula</b>                              | C <sub>6</sub> H <sub>12</sub> N <sub>3</sub> P S                                                                                                                                                                                                                  |
| <b>Supplier</b>                                       | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608                                                       |
| <b>Emergency Telephone Number</b>                     | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| <b>E-mail address</b>                                 | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| <b>Recommended Use</b><br><b>Uses advised against</b> | Laboratory chemicals.<br>No Information available                                                                                                                                                                                                                  |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                                |                            |                       |
|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------------------|
| <b>Physical State</b><br>Powder Solid                                                                                                          | <b>Appearance</b><br>White | <b>Odor</b><br>Slight |
| <b>Emergency Overview</b><br>Fatal if swallowed. May cause genetic defects. May cause cancer. May form combustible dust concentrations in air. |                            |                       |

#### Classification of the substance or mixture

|                        |             |
|------------------------|-------------|
| Acute Oral Toxicity    | Category 2  |
| Germ Cell Mutagenicity | Category 1A |
| Carcinogenicity        | Category 1A |

#### Label Elements



Signal Word

Danger

## Tri(1-aziridinyl)phosphine sulfide

**Hazard Statements**

H300 - Fatal if swallowed  
H340 - May cause genetic defects  
H350 - May cause cancer

**Precautionary Statements****Prevention**

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P330 - Rinse mouth  
P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor  
P308 + P313 - IF exposed or concerned: Get medical advice/attention

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified. May form combustible dust concentrations in air.

**Health Hazards**

Very toxic if swallowed. May cause genetic defects. May cause cancer.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

May form explosible dust-air mixture if dispersed. This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component                           | CAS No  | Weight % |
|-------------------------------------|---------|----------|
| Tris(1-aziridinyl)phosphine sulfide | 52-24-4 | > 98     |

**SECTION 4. FIRST AID MEASURES****Eye Contact**

Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

**Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Immediate medical attention is required.

**Inhalation**

Remove from exposure, lie down. Remove to fresh air. If not breathing, give artificial respiration. Immediate medical attention is required.

**Ingestion**

Call a physician immediately. Clean mouth with water.

**Most important symptoms and effects**

No information available.

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Fine dust dispersed in air may ignite.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation.

**Environmental Precautions**

See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Wear self-contained breathing apparatus and protective suit. Sweep up and shovel into suitable containers for disposal. Do not let this chemical enter the environment.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Do not breathe dust. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance. Handle product only in closed system or provide appropriate exhaust ventilation.

**Storage**

Keep in a dry place. Keep container tightly closed. Keep locked up. Keep refrigerated.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls**

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Neoprene       | recommendations   |                 |             |                       |
| Natural rubber |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES**

|                                 |                               |                                          |
|---------------------------------|-------------------------------|------------------------------------------|
| <b>Appearance</b>               | White                         |                                          |
| <b>Physical State</b>           | Powder Solid                  |                                          |
| <b>Odor</b>                     | Slight                        |                                          |
| <b>Odor Threshold</b>           | No data available             |                                          |
| <b>pH</b>                       | No information available      |                                          |
| <b>Melting Point/Range</b>      | 54 - 57 °C / 129.2 - 134.6 °F |                                          |
| <b>Softening Point</b>          | No data available             |                                          |
| <b>Boiling Point/Range</b>      | No information available      |                                          |
| <b>Flash Point</b>              | No information available      | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>         | Not applicable                | Solid                                    |
| <b>Flammability (solid,gas)</b> | No information available      |                                          |
| <b>Explosion Limits</b>         | No data available             |                                          |

## Tri(1-aziridinyl)phosphine sulfide

|                                         |                          |       |
|-----------------------------------------|--------------------------|-------|
| Vapor Pressure                          | No data available        |       |
| Vapor Density                           | Not applicable           | Solid |
| Specific Gravity / Density              | No data available        |       |
| Bulk Density                            | No data available        |       |
| Water Solubility                        | 19g/100mL (25°C)         |       |
| Solubility in other solvents            | No information available |       |
| Partition Coefficient (n-octanol/water) |                          |       |
| Autoignition Temperature                | No data available        |       |
| Decomposition Temperature               | No data available        |       |
| Viscosity                               | Not applicable           | Solid |
| Explosive Properties                    | No information available |       |
| Oxidizing Properties                    | No information available |       |

|                   |               |
|-------------------|---------------|
| Molecular Formula | C6 H12 N3 P S |
| Molecular Weight  | 189.23        |

## SECTION 10. STABILITY AND REACTIVITY

|                          |                                 |
|--------------------------|---------------------------------|
| Stability                | Stable under normal conditions. |
| Hazardous Reactions      | No information available.       |
| Hazardous Polymerization | No information available.       |
| Conditions to Avoid      | Incompatible products.          |
| Materials to avoid       | Strong oxidizing agents.        |

**Hazardous Decomposition Products** Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides. Oxides of phosphorus. Burning produces obnoxious and toxic fumes.

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component                           | LD50 Oral               | LD50 Dermal | LC50 Inhalation |
|-------------------------------------|-------------------------|-------------|-----------------|
| Tris(1-aziridinyl)phosphine sulfide | LD50 = 23 mg/kg ( Rat ) |             |                 |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

## (d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

(e) germ cell mutagenicity; Category 1A

(f) carcinogenicity; Category 1A

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component                           | EU | UK | Germany | IARC    |
|-------------------------------------|----|----|---------|---------|
| Tris(1-aziridinyl)phosphine sulfide |    |    |         | Group 1 |

|                                           |                                                                |
|-------------------------------------------|----------------------------------------------------------------|
| (g) reproductive toxicity;                | No data available                                              |
| (h) STOT-single exposure;                 | No data available                                              |
| (i) STOT-repeated exposure;               | No data available                                              |
| Target Organs                             | No information available.                                      |
| (j) aspiration hazard;                    | Not applicable<br>Solid                                        |
| Other Adverse Effects                     | The toxicological properties have not been fully investigated. |
| Symptoms / effects,both acute and delayed | No information available                                       |

**SECTION 12. ECOLOGICAL INFORMATION**

|                                                                                              |                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ecotoxicity effects                                                                          | Do not empty into drains.                                                                                                                                                                                     |
| Persistence and Degradability<br>Persistence                                                 | Soluble in water, Persistence is unlikely, based on information available.                                                                                                                                    |
| Bioaccumulative Potential                                                                    | Bioaccumulation is unlikely                                                                                                                                                                                   |
| Mobility in soil                                                                             | The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils                                                     |
| Endocrine Disruptor Information<br>Persistent Organic Pollutant<br>Ozone Depletion Potential | This product does not contain any known or suspected endocrine disruptors<br>This product does not contain any known or suspected substance<br>This product does not contain any known or suspected substance |

**SECTION 13. DISPOSAL CONSIDERATIONS**

|                                     |                                                                                                                                                                        |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste from Residues/Unused Products | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations. |
| Contaminated Packaging              | Dispose of this container to hazardous or special waste collection point.                                                                                              |
| Other Information                   | Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.                                          |

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

|                      |                              |
|----------------------|------------------------------|
| UN-No                | UN2811                       |
| Proper Shipping Name | Toxic solid, organic, n.o.s. |
| Hazard Class         | 6.1                          |
| Packing Group        | II                           |

## Tri(1-aziridinyl)phosphine sulfide

**IMDG/IMO**

UN-No UN2811  
Proper Shipping Name Toxic solid, organic, n.o.s.  
Hazard Class 6.1  
Packing Group II

**IATA**

UN-No UN2811  
Proper Shipping Name TOXIC SOLID, ORGANIC, N.O.S.\*  
Hazard Class 6.1  
Packing Group II

Special Precautions for User No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component                           | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL |
|-------------------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|------|
| Tris(1-aziridinyl)phosphine sulfide | -                                                   | -                                       | X    | -     | 200-135-7 | -    | -   | -     | X    | X    | -    | -    |

**National Regulations****SECTION 16. OTHER INFORMATION**

Prepared By Health, Safety and Environmental Department  
Revision Date 25-Apr-2024  
Revision Summary New emergency telephone response service provider.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Legend**

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

## Tri(1-aziridinyl)phosphine sulfide

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**WEL** - Workplace Exposure Limit**ACGIH** - American Conference of Governmental Industrial Hygienists**DNEL** - Derived No Effect Level**RPE** - Respiratory Protective Equipment**LC50** - Lethal Concentration 50%**NOEC** - No Observed Effect Concentration**PBT** - Persistent, Bioaccumulative, Toxic**TWA** - Time Weighted Average**IARC** - International Agency for Research on Cancer**PNEC** - Predicted No Effect Concentration**LD50** - Lethal Dose 50%**EC50** - Effective Concentration 50%**POW** - Partition coefficient Octanol:Water**vPvB** - very Persistent, very Bioaccumulative**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road**OECD** - Organisation for Economic Co-operation and Development**BCF** - Bioconcentration factor**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code**MARPOL** - International Convention for the Prevention of Pollution from Ships**ATE** - Acute Toxicity Estimate**VOC** - (Volatile Organic Compound)**Key literature references and sources for data**<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

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**End of Safety Data Sheet**